

LAPPEENRANTA HIEKKAPAKKA, Finland

WMO: 029190

Lat: 61.198N

Lon: 28.473E

Elev: 253

StdP: 14.56

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-11.1	-22.4	1.6	-17.4	-15.8	2.3	-11.0	29.1	29.9	27.2	28.6	2.4	280	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	78.5	65.4	75.2	64.1	72.2	62.2	67.5	74.9	65.6	73.0	63.6	70.5	7.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.9	92.9	72.0	62.5	85.5	69.3	60.5	79.4	67.3	32.1	74.7	30.5	73.2	29.0	70.6	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.3	23.6	21.2	DB	-19.2	82.2	10.5	4.5	-26.8	85.5	-32.9	88.1	-38.8	90.7	-46.4	93.9	(4)
(5)			WB	-19.5	69.1	10.4	2.5	-27.0	70.9	-33.1	72.4	-38.9	73.8	-46.4	75.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	17.5	17.2	25.6	35.4	47.2	58.3	62.2	60.3	51.3	40.1	29.0	23.5
	DBStd	18.36	12.49	13.92	8.79	7.58	9.07	6.56	6.25	6.25	7.13	6.76	8.85	12.04
	HDD50	5134	1008	919	756	440	165	8	1	5	71	310	629	822
	HDD65	9538	1473	1339	1221	887	553	218	131	169	410	771	1079	1287
	CDD50	1157	0	0	0	2	78	258	380	324	111	4	0	0
	CDD65	87	0	0	0	0	1	18	45	23	0	0	0	0
	CDH74	404	0	0	0	0	9	103	219	73	0	0	0	0
	CDH80	47	0	0	0	0	0	19	26	2	0	0	0	0
Wind	WSAvg	9.8	9.2	9.0	8.7	8.1	8.6	10.0	8.8	9.8	10.3	11.9	11.2	11.4
Precipitation	PrecAvg	25.2	1.9	1.5	1.4	1.2	1.7	2.4	3.0	2.9	2.3	2.5	2.4	2.1
	PrecMax	31.2	3.1	3.6	2.6	2.5	5.6	3.6	5.8	5.3	5.2	5.8	5.0	4.5
	PrecMin	19.4	0.4	0.2	0.3	0.4	0.3	0.7	0.7	0.6	0.7	0.6	0.4	0.8
	PrecStd	2.8	0.7	0.8	0.6	0.5	1.1	0.7	1.2	1.4	1.3	1.1	1.0	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.6	37.7	44.3	62.7	73.9	83.6	83.1	79.5	69.9	55.4	47.7	44.0
		MCWB	35.1	35.7	36.9	51.2	60.3	65.4	67.3	66.0	60.9	51.5	45.6	42.3
	2%	DB	35.3	35.9	40.2	54.7	69.6	76.4	79.3	75.8	65.3	52.9	44.8	40.0
		MCWB	34.3	34.7	35.0	45.0	57.0	63.2	66.0	64.4	58.7	50.4	43.8	38.4
	5%	DB	33.7	34.2	37.6	50.0	65.5	72.8	76.2	72.9	63.1	51.2	42.0	37.3
		MCWB	32.8	33.3	33.8	42.4	54.8	61.2	65.4	62.8	57.8	49.1	41.1	36.1
	10%	DB	32.4	33.0	35.7	45.9	61.3	69.0	72.9	70.0	60.8	49.3	39.9	35.3
		MCWB	31.8	32.3	32.9	40.0	52.7	59.7	63.6	61.8	56.8	47.3	39.0	34.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	36.3	38.2	51.0	62.4	67.2	70.4	69.3	62.4	52.6	45.9	42.8
		MCDB	36.3	37.1	41.9	62.1	72.1	77.5	77.6	75.4	66.6	54.3	47.1	44.0
	2%	WB	34.4	35.0	36.1	46.3	59.0	64.9	68.5	66.6	60.3	51.1	43.8	38.5
		MCDB	35.0	35.9	38.7	52.6	65.9	73.2	75.9	73.0	63.4	52.5	44.8	39.7
	5%	WB	33.1	33.5	34.6	43.2	56.6	63.0	66.6	64.2	58.8	49.6	41.3	36.1
		MCDB	33.7	34.0	36.8	49.0	63.4	70.5	74.0	70.8	61.9	51.1	41.9	36.9
	10%	WB	32.0	32.4	33.4	40.5	53.4	60.9	64.6	62.3	57.1	47.6	39.0	34.6
		MCDB	32.5	33.0	35.2	45.3	60.8	68.0	71.7	68.4	60.5	49.2	39.9	35.3
Mean Daily Temperature Range	5% DB	MDBR	10.0	11.3	13.0	13.7	13.9	13.1	12.6	12.3	9.3	6.5	6.5	7.7
		MCDBR	7.0	7.7	13.7	19.1	20.4	19.3	17.0	16.8	11.5	7.0	6.4	5.6
		MCWBR	7.1	7.2	10.0	11.5	10.7	9.4	7.6	7.8	6.6	5.6	6.3	5.6
	5% WB	MCDBR	6.8	6.2	10.2	17.5	16.8	15.9	14.0	11.6	10.0	6.6	6.7	5.4
		MCWBR	6.8	6.0	7.8	11.2	9.4	8.7	7.3	6.3	6.3	5.7	6.6	5.5
Clear-Sky Solar Irradiance	taub		0.255	0.298	0.324	0.363	0.358	0.355	0.380	0.366	0.342	0.324	0.278	N/A
	taud		2.066	2.124	2.200	2.336	2.431	2.470	2.418	2.444	2.502	2.401	2.155	N/A
	Ebn at Noon		143	209	249	263	274	277	266	260	247	206	134	N/A
	Edh at Noon		13	23	30	32	32	31	32	29	23	18	12	N/A
All-Sky Solar Radiation	RadAvg		76	254	716	1166	1601	1686	1645	1268	738	297	80	39
	RadStd		14	60	99	126	187	110	155	148	81	42	13	5

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.97	N/A	N/A	N/A	N/A	N/A	-342	-340	N/A	N/A

Nomenclature: See separate page